Few-shot Learning with Siamese Networks

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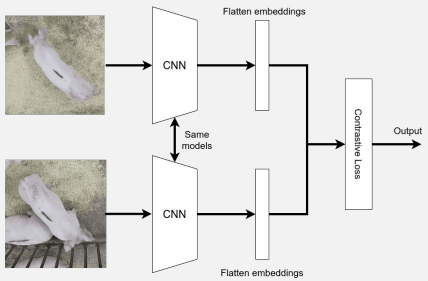
Siamese network

Few-shot learning = learning from few labeled examples



N-way = # classes in support set
K-shot = # examples / class

Generalize to unseen classes with few examples
→ Inspired by human learning

Data

Dataset mini-ImageNet

Data loaders

1. Pairs (pos. / neg.)
2. Triplets (A, P, N)

Data split

Pair

Triplet

Model optimization

Contrastive loss

$$L = \frac{1}{2} \cdot y \cdot d^2 + \frac{1}{2} \cdot (1 - y) \cdot \max(0, m - d)^2$$

d = distance; **y** = label (pos. pair: y = 1, neg. pair = 0); **m** = margin

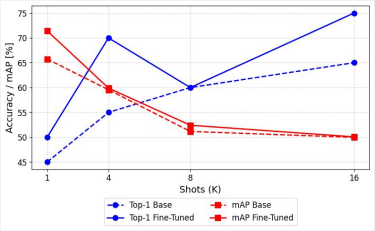
Triplet loss

$$L = \max(0, d(A, P) - d(A, N) + m)$$

d = distance; **A** = anchor; **P/N** = pos./neg. image; **m** = margin

Custom CNN

Contrastive vs Fine-Tuned Contrastive



Fine-tuning the contrastive model boosts Top-1 accuracy and improves embeddings for few-shot tasks

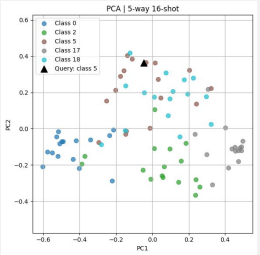
Contrastive vs Triplet Learning

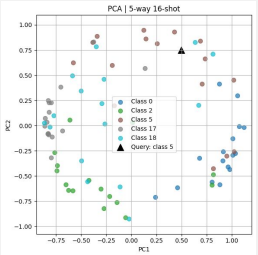
Quantitative results

N-way K-shot	Top-1 accuracy [%]	Top-5 accuracy [%]	Mean Average Precision [%]
5-way 1-shot	45 / 45	100 / 100	65 / 65
5-way 4-shot	**55** / 50	90 / 80	**59** / 51
5-way 8-shot	**60** / 45	90 / 90	**51** / 48
5-way 16-shot	**65** / 50	90 / 85	**50** / 43

red: **contrastive model**; black: **triplet model**

PCA (C=5) **Contrastive** | **Triplet**





Contrastive loss improves few-shot accuracy; triplet loss primarily yields clustering with limited classification gains

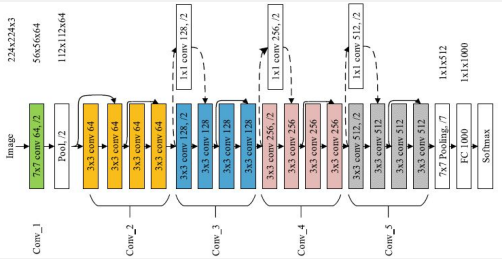
Grad-CAM **Contrastive** | **Triplet**

ResNet-18

Dataset

Places365 (scene recognition)

Model architecture



ResNet-18

```
Custom layers summary:
----------------------------------------------------------------
        Layer (type)               Output Shape
================================================================
            Conv2d-1             [-1, 512, 7, 7]
       BatchNorm2d-2             [-1, 512, 7, 7]
              ReLU-3             [-1, 512, 7, 7]
            Dropout-4             [-1, 512, 7, 7]
 AdaptiveAvgPool2d-5             [-1, 512, 1, 1]
            Flatten-6                  [-1, 512]
             Linear-7                  [-1, 256]
       BatchNorm1d-8                  [-1, 256]
              ReLU-9                  [-1, 256]
================================================================
Total params: 2,492,672
Trainable params: 2,492,672
Non-trainable params: 0
```

Custom layers

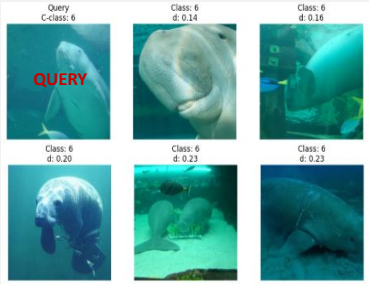
Evaluation

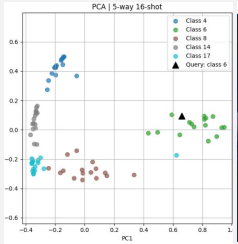
Quantitative results

N-way K-shot	Top-1 accuracy [%]	Top-5 accuracy [%]	Mean Average Precision [%]
5-way 1-shot	60 / 69	100 / 100	76 / 81
5-way 4-shot	65 / 69	100 / 94	67 / 70
5-way 8-shot	65 / 81	90 / 100	67 / 76
5-way 16-shot	70 / 81	95 / 94	65 / 65

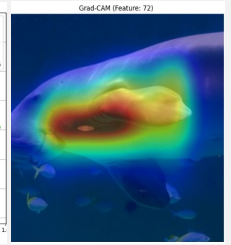
red: **contrastive model**; black: **triplet model**

Qualitative results (contrastive)





PCA



Grad-CAM